Refactoring the GRC pipeline

What changed in RP7, and why

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The pipeline had drifted into 22 do-files we could not audit

By April 2026, three structural problems had compounded:

- Twenty-two do-files in `scripts/`, with the experience and birth tables each living in their own per-country file, even though the regression call was nearly identical
- A parallel folder `ReplicationPackage6 - real values/` with a copy of every script, deflating consumption and re-running everything—two trees that drifted whenever either one was edited
- Ster filenames colliding across sections, and a few exceeding Stata's 32-character internal name limit, which surfaced as `r(7) invalid name` deep in the call chain

None of this changes the published numbers. All of it makes the pipeline harder to maintain, to onboard a coauthor onto, and to smoke-test end-to-end.

We collapsed to 13 do-files via three program-level moves

Before (RP6, 22 files)

- `5_GrRC.do`
- `6_GrRC_NonAg.do`
- `8_GrRC_hukou.do`
- `10_GrRC_experience.do`
- `11_GrRC_max_experience.do`
- `12_GrRC_experience_share.do`
- `13_GrRC_max_experience_share.do`
- `14_GrRC_NonAg_experience.do`
- `15_GrRC_birth.do`
- `16_heterogeneity_tables.do`
- ...

After (RP7, 13 files)

- `5_GrRC.do`
- `6_GrRC_NonAg.do`
- `8_GrRC_hukou.do`
- `GRC_extras.do` (experience family dispatcher)
- `make_tables.do`, `make_figures.do`
- `16_heterogeneity_tables.do`

Net: −9 files, same numbers

M1 + M2: the experience and birth families no longer have their own scripts

The five experience-family scripts (10, 11, 12, 13, 14) and `15_GrRC_birth.do` all did the same thing: take the GRC regression and add one extra regressor (experience, max experience, share of urban experience, max share, birth dummy).

New machinery in `0_programs.do`

- `run_grc_with_extra_regressor`—one program, takes `regressor()` and `spec3()` arguments
- `GRC_extras.do`—thin dispatcher; one cell per (country, spec3, regressor) triple
- Sters land at `grc_<country>_<spec3>_<fam>_<col>.ster` (M11 naming)

The `extras_tex_table` program reads those sters back and emits the family-specific .tex tables. The Tier 2 byte-identity check confirms the .tex bodies match what RP6 produced, modulo the two whitelisted refactor patterns (Phase 1b).

M3: one `grc_tex_table_trend` replaces four

`grc_tex_table_trend`, `grc_tex_table_trend_exp`, `grc_tex_table_trend_birth`, and `grc_tex_table_trend_hukou` were 95% the same code with different stored-name lookups and slightly different headers.

The unified program now takes:

- `spec()`—`cuu`, `cub`, `iuu`, `cnu`, or empty for hukou (which encodes the spec in `country_short`)
- `covs2set()`—defaults to `c0 ct c1 c2 ca`, overrideable for the experience family
- `filename()`, `keep()`, `varlabel()`, `postfoot()`, `coeflabels()`, `textdepvar()`

A side effect of this consolidation surfaced last night: each call stores roughly fifteen estimates in memory, and Stata's `est store` limit is 300, so by the 21st cell the namespace overflowed. The fix was a one-line `est drop _all` at the program's tail (commit `b7b8313`).

M4: one switch handles nominal and real values

Before: two parallel Dropbox folders. `ReplicationPackage6/` ran the pipeline on nominal values; `ReplicationPackage6 - real values/` ran the same pipeline on consumption deflated to a base year. Every edit had to be propagated to both trees by hand.

After: a single `values(nominal|real)` global in `0_path_config.do`.

- `nominal` (default): output filenames unchanged, byte-identical to pre-M4 reference
- `real`: appends `_r` suffix (`$vsfx`) to every ster, table, and figure—the two modes coexist in one `output/` tree
- `RP7/data_real/` is a new directory junction onto `ReplicationPackage6 - real values/data/`; gitignored, set up per machine

A four-test harness (`tests/verify_M4_values_switch.do`) confirmed nominal-mode output is bit-identical to RP6's pre-M4 sters on `grc_IDN_cub_c0`.

M11: ster filenames are now unique and 32-char-safe

The old naming scheme had two flaws:

- Names like `grc_robust_vv_TZA_onestep_covs_trend` (34 chars) tripped Stata's 32-character limit on `_est_<name>`, throwing `r(7)` far from the offending `eststo`
- Sections 5, 6, and 8 stored sters under names that could collide in the same session

New convention (`RP7/scripts/STER_NAMING.md`):

`grc_<country>_<spec3>_<covs2>[_<sfx1>]`

- `country`: `IDN`, `CHN`, `TZA`, or `CHN_hukou_<subgroup>`
- `spec3`: `cuu`, `cub`, `iuu`, `cnu`
- `covs2`: `c0`, `ct`, `c1`, `c2`, `ca`
- `sfx1`: `n` (never), `a` (always), `d` (delta), `g` (average)

No name in production exceeds 27 characters.

M9 + M10: smokes survive interruptions, and every fit times itself

Tier 3 (the full master pipeline) runs roughly thirty wall-clock hours. Two changes make that survivable.

M9—per-fit timer. `run_grc` and its variants record elapsed seconds in `e(elapsed_s)`. We can sort sters by run time, spot blown-up cells, and budget the next smoke.

M10—resume-on-interrupt. A new global `$skip_if_exists` tells `run_grc` to skip cells whose ster is already on disk. When the 2026-05-01 OOM killed Tier 3 #5 mid-run, relaunching with `skip_if_exists=1` carried 1,365 sters forward and re-ran only the missing 140. Tier 3 #6 closed cleanly at 06:50 on 2026-05-03 with 1,505 sters.

Tables: $\bar{\Delta}$ replaces “Average Δ”, envelope moves to the paper

Phase 1b separated paper-side concerns from code-side concerns.

What lives where now

- Paper preamble: caption text, `\label{}`, and table-notes, wrapped in `\GRCtable{}` and `\GRChukoutable{}` macros
- Stata output: tabular-only fragment plus the indicator rows (Time FE, Covariates)

Cosmetic changes in the fragments themselves

- Average Δ row reads $\bar{\Delta}$ (commit `5e2277c`)
- esttab's blank tabular rows stripped via `removeStringFromTex` (commit `f68892e`)

The Tier 2 harness whitelists exactly these two patterns (`LABEL_FLIP`, `BLANK_ROW`); every other byte-level diff is `UNEXPECTED` and blocks the merge.

Verification holds at every layer

Tier	Scope	Result
Tier 1 (static)	call-site audit, ster-name length, lint	5/6 PASS
Tier 2 (table body)	53 reference `.tex` files, classified diff	53 expected, 0 unexpected, 0 missing
Tier 3 (full smoke)	full master pipeline, convergence, ster counts	1,505 sters, exit 0
M4 verify	4-test driver on `grc_IDN_cub_c0`	4/4 PASS
M4 mu-loop	bit-comparison on 4 production cells	PASS

The one Tier 1 check that does not PASS flagged 75 pre-M11 orphan sters in `RP7/output/`. Those are leftover names from earlier in the refactor; they live alongside the new ones, do not feed anything, and will be cleaned up when the branch lands.

Tier 2 byte-identity totals: `LABEL_FLIP=53`, `BLANK_ROW=54`, `ADDLINESPACE=0`, `UNEXPECTED=0`.

How to run the pipeline (cheat sheet)

Full pipeline, nominal values

```
cd RP7/scripts && stata-mp -b do 0_master.do
```

Single section (re-runnable, e.g. just the main GRC regressions)

```
cd RP7/scripts && stata-mp -b do 5_GrRC.do
```

Tables only—regenerate every `.tex` from existing sters, no GMM

```
cd RP7/scripts && stata-mp -b do _smoke_tables_only.do
```

Tier 2 byte-identity check vs. the snapshot

```
python tests/tier2_table_diff.py
```

Real-values mode: edit `global values "real"` in `0_path_config.do`, then re-run.

Where things live in `RP7/`

- `RP7/scripts/`—13 do-files; all authoritative edits live here
- `RP7/data/`—junctioned to the shared Dropbox `ReplicationPackage6/data/`; raw `.dta` files are immutable
- `RP7/data_real/`—junctioned to `ReplicationPackage6 - real values/data/`; only consulted when `values=real`
- `RP7/output/`—our reruns land here. Subdirs:
    - `tables/`—72 `.tex` files (tracked in git)
    - `figures/`—PDF + PNG (tracked)
    - `*.ster`—estimation results (gitignored; regeneratable from code)
- `tests/`—verification harnesses, including `tier2_table_diff.py` and `verify_M4_*.do`
- `quality_reports/specs/`, `docs/plans/`, `quality_reports/session_logs/`—per-decision documentation, including the spec at `2026-04-24_grc-pipeline-refactor.md`

Open items and handoff

- PR #3 is open against `main`: 82 commits, 245 files changed
- Tier 2 reference snapshot covers 53 of the 72 tables the pipeline now produces. The 12 hukou, 1 IDN nonag, and 6 heterogeneity tables sit outside the snapshot—a pre-existing gap, not a refactor regression. Worth refreshing post-merge if we want a tighter Tier 2 net
- Paper-side `\GRCvaluesfx` toggle macro is deferred. It will go into Overleaf `preamble.tex` once we decide whether real-values is a replacement for the nominal-values tables, or a robustness check that lives alongside them
- When this branch lands on `main`, copy `RP7/{scripts,output}/` into Dropbox as `ReplicationPackage7/`. The Dropbox-side handoff is the only piece that touches the team's shared workspace

What this buys us

- One source of truth for the regression code (no more script fork to sync)
- One source of truth for the table envelope (in the paper preamble, where it belongs)
- Ster filenames a coauthor can read, type, and grep for
- A Tier 2 byte-identity check that tells us within seconds whether a code change altered a published number
- Smokes that resume after a hardware hiccup, with timing data per fit

The published numbers are unchanged. The pipeline is now small enough to hold in one head, and audited well enough that any further change leaves a clear paper trail.